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REVIEW ARTICLE

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Inventory rationing with critical level policies: a structured literature review

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ABSTRACT

Differentiating service for various customer classes is a critical challenge in modern supply chains, particularly in sectors like e-commerce, healthcare, and industries reliant on critical spare parts. Inventory rationing with critical level policies is a key mechanism to address this challenge. This study presents a comprehensive, structured literature review of 128 studies on this topic. We classify existing research based on problem characteristics, such as the distinction between capacitated and non-capacitated systems, and policy structures, which we categorise as static, dynamic, or state-dependent. Our findings show that lead time uncertainty is an important factor determining whether the optimal policy is dynamic or state-dependent, and that simpler static policies often perform remarkably close to complex optimal ones, offering a highly practical solution for managers. Based on this synthesis, we identify research gaps, such as the study of rationing policies for capacitated systems with service level constraints and real-world applications like healthcare, where the differentiation between critical and non-critical supplies, taking into account product substitutions, is vital.

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1. Introduction

Supply chains delivering service to various types of customers may need to consider different requirements for different customers. While some customers might be willing to wait for a product, others expect prompt and reliable deliveries. This variety in requirements between customer segments served by the same company is typical in e-commerce (Nguyen et al. 2019) and omnichannel retail (Berling, Johansson, and Marklund 2023). It may also be present, for instance, in sectors using critical components in their operations (e.g. critical spare parts (Kouki and Larsen 2021) in the aviation or petrochemical sector (Teunter and Klein Haneveld 2008)), in sectors with scarce resources such as healthcare (e.g. blood donations (Civelek, Karaesmen, and Scheller-Wolf 2015) or vaccines (Samii et al. 2012)), or by the simple (un)willingness of a market segment to wait for a product. For example, de Leeuw, Holweg, and Williams (2011) identify that people are not willing to wait for the delivery of their new vehicle in the USA, whereas in Europe this situation is different.

One way of providing a higher service level to some customers is to reserve available on-hand inventory for higher priority customers only, which is not to be released to lower priority customers (Teunter and Klein Haneveld 2008). This is done because the total demand requirements could be higher than the available inventory before the next replenishment comes in, thus jeopardising high priority customers over lower priority ones when there is no differentiation between customer types.

The literature has shown that this inventory reservation challenge can be optimally addressed for some settings by critical level policies. These are policies in which on-hand inventory is reserved for (future) high-priority demand whenever the on-hand inventory is less than a critical threshold level c (Nahmias and Demmy 1981), effectively withholding on-hand inventory from low-priority demand. According to this rule, if the inventory on-hand at the central warehouse/factory I_0 is smaller than the threshold level (i.e. $I_0 < c$), then all incoming lower priority demand will be directly backordered (or lost) until the next central stock-point replenishment arrives, while all high priority demand will be immediately fulfilled to the extent possible. If the number of customer classes N is greater than two, then there are typically N-1 critical levels c_i defining the inventory threshold levels in which all classes below class i are backordered (or lost) and all classes above and equal to class i are fulfilled. This is assuming that customer classes are ordered from 1 to N in decreasing order of

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priority, i.e. class 1 is the highest priority class and has the highest shortage penalty. The main objective of this mechanism is to prevent a possible, but uncertain, higher unrealised penalty at a decision moment by incurring a certain, lower penalty since there could not be enough stock on hand to fulfil all incoming demand. Thus, inventory rationing studies assume that $p_1 > p_2 > \ldots > p_N$, being that p_i is the shortage penalty cost (or priority) for customer class i, for all $i \in \{1, 2, \ldots N\}$, N being the number of customer classes.

While the work by Kleijn and Dekker (1999) provided a first overview of the field, research on inventory rationing has since evolved to address more complex operational environments. The original classification, focused on the number of customer classes and the type of inventory policy, is no longer sufficient to capture the characteristics of the current body of literature. Since their review, research has expanded in terms of the inventory system setting, exploring, for example, capacitated production-inventory systems with component commonalities and profit-maximising problems with features like price-dependent demand. Although subsequent studies have offered valuable categorizations (see, e.g. Arslan, Graves, and Roemer 2007; Goedhart, Haijema, and Akkerman 2022; Möllering and Thonemann 2008; Teunter and Klein Haneveld 2008), they have primarily done so to position their own work rather than providing a systematic overview of the entire field. Therefore, an extensive, updated classification is needed to provide an up-to-date review of research on inventory rationing with critical level policies.

This study fills that gap by systematically reviewing 128 studies on inventory rationing policies with critical levels. To structure this review, we address the following research questions:

> RQ1. What are the main problem characteristics and modelling assumptions that define the research on inventory rationing with critical level policies?
>
> RQ2. How are inventory rationing policies structured, and under which problem settings have these policy structures been shown to be optimal?
>
> RQ3. Based on the existing body of literature, what are the promising directions for future research?

Our contribution is twofold: (1) we provide a structured classification of main problem assumptions and characteristics, including demand processes, lead-time considerations, service level constraints, and capacity considerations; and (2) we analyse how different rationing policies are structured, highlighting their optimality under various operating conditions. By integrating these insights, we present a comprehensive classification of research on inventory rationing, considering a broader range of problem settings, policy structures, and solution approaches. This classification helps identify research gaps and future directions while also bridging theoretical insights with practical applications.

Since this review intentionally focuses on one specific mechanism for inventory control, it is positioned between more general structured overviews of inventory management systems such as the ones from de Kok et al. (2018) and Silver, Pyke, and Thomas (2016), who cover a wide range of problems and approaches, and reviews concerned with specific sectors, e.g. spare parts sector (Topan et al. 2020) or assemble-to-order systems (Atan et al. 2017). While the specialised reviews by Topan et al. (2020) and Atan et al. (2017) consider rationing policies just as part of the many operational policies applied in some specific settings, our work provides a cohesive analysis of the rationing policy's underlying structure and optimality conditions across different problem application areas. This focused approach is necessary to identify fundamental principles of policy design that apply beyond any single application area.

The remainder of this study is organised as follows: Section 2 explains the methodology used to conduct this investigation. A structured overview of the literature on inventory rationing is presented in Section 3. Section 4 focuses on the discussion and future research opportunities, and the conclusions of this study are presented in Section 5.

2. Methodology

For this literature review, we followed the steps of Denyer and Tranfield (2009). First, we explored the literature of this field by reading several studies while identifying relevant keywords to use in a subsequent search. Since our classification significantly expands upon the framework established by Kleijn and Dekker (1999), we do not impose a specific time period as a constraint, as is common in literature review updates. Instead, we incorporate all relevant studies, including those covered in Kleijn and Dekker's review, to provide a more comprehensive and inclusive classification of the field without a lower cutoff date. We only consider peer-reviewed and English-written articles published in journals. Second, we formulated three search strings for a title-based search in the Web of Science database. The search strings combined terms related to the general mechanism (e.g. 'inventory rationing', 'stock allocation') with terms for the specific policy type (e.g. 'critical level policy', 'rationing level'). The full, detailed search strings are provided in Appendix A. The initial search was conducted in February 2022, and it was updated in December 2024 to capture the most recent publications since the original search. The purpose

of this update was to ensure the review's comprehensiveness at the time of publication. We selected relevant articles based on title and abstract screening, focusing on the discussion of inventory rationing policies. Third, we removed duplicates and closely examined the remaining articles.

Since there is an overlap between shortage rationing studies and inventory rationing (critical level) studies, we carefully screened the text of the studies to select only those studies dealing with inventory rationing policies using critical level policies. While the term 'rationing' has also been used to refer to the problem of *shortage rationing* (Diks and de Kok 1996; Lagodimos and Koukoumialos 2008) where one needs to decide exactly how much stock to allocate to each stockpoint in a divergent network when the available stock is not sufficient to completely fulfil all realised or expected demands, here we focus only on policies where inventory is *withheld* from low-priority realised demand.

We excluded studies related exclusively to capacity or demand allocation/rationing (see, e.g. Benjaafar, ElHafsi, and de Véricourt 2004; Hung and Lee 2010) when no consideration was made to inventory rationing with critical level policies. While these problems share similarities with critical level policies in terms of solution procedures and policy structures, they fundamentally differ because inventory is not depleted and carried over single or multiple periods. Additionally, we excluded studies that treat critical level policies merely as fixed parameters within a model, without explicitly estimating or optimising the critical level (see, e.g. Escalona et al. 2018, 2015).

To answer our research questions, we developed a classification framework that builds on established typologies in the inventory management literature. To address RQ1, we adapted the typology from de Kok et al. (2018) to classify the problem characteristics and modelling assumptions in the reviewed studies. To answer RQ2, we extended the policy classifications from Möllering and Thonemann (2008) and Teunter and Klein Haneveld (2008) to analyse the different rationing policy structures. Finally, for RQ3, we synthesised these results to identify trends and limitations, which we then contrasted with current operational challenges to identify research gaps. The results pertaining to RQ1 and RQ2 are shown in the next section; whereas RQ3 is addressed in Section 4.1.

3. Structured analysis of the literature

From a full screening of all the studies, a general structure of the field emerged, where we identified a number of attributes common to all 128 studies included in this review. These attributes were further converted into parameters that could help understand the structure of the field in a straightforward manner. Table 1 provides a summary of the common attributes identified throughout the reviewed studies as well as their parametrization.

The first set of attributes describes the methodological approach of the studies. Thus, the modelling methods and goals have been identified. The second group of attributes identified in the studies is the characteristics of the inventory rationing problems, which describe the problem setting. Typical problem attributes include the consideration of capacitated or non-capacitated replenishment/ordering processes, distributional assumptions for replenishment lead times and customer demand, and the number of customer classes. In addition, the response of customer arrivals under stockouts and service level constraints were also included as attributes of the problem setting, whereas inventory replenishment/ordering policies and single-period or multiperiod settings were also considered, as well as the costs and the performance objectives considered in the studies. Finally, we classify the inventory rationing policies according to their structure and indicate which of these policies was found to be optimal.

In the following sections, we provide a brief explanation of each attribute as well as an overview of the literature in terms of problem characterisation, classification and structure of rationing policies, and methodological approaches used.

3.1. Characterisation of the problems

3.1.1. Supply considerations

Inventory rationing studies deal with both non-capacitated and capacitated replenishment (ordering) processes. Capacitated models consider the replenishment lead time as the result of the stochastic process of a capacity-constrained resource system, e.g. an M/M/1 queue, effectively modelling production-inventory systems (classified as production systems by Teunter and Klein Haneveld (2008)). On the other hand, non-capacitated models do not consider capacity-constrained resources supplying the network and, as such, regard the lead time as a variable that is independent of the size of the order, modelling the lead time as a probabilistic function exogenous to the model, e.g. a normally distributed lead time. Because of the added complexity of modelling capacitated systems, the majority of studies on inventory rationing policies (83 studies) have considered non-capacitated models while a smaller set (45) considers capacitated models.

Bar a few exceptions, all studies considered a single-echelon supply chain network fulfilling the demands of different incoming customer classes, irrespective of whether or not the studies considered a capacitated

Table 1. Summary of attributes identified in the studies.

Parameter	Values	Number of studies
Methodological approach		
Modelling method[1]	Exact	104
	Approximate exact model (truncation)[2]	25
	Approximate	35
	Simulation	38
	Numerical experiments[3]	95
Research goal	Optimisation	112
	Performance evaluation	16
Problem characterisation		
Supply capacity	Non-capacitated	83
	Capacitated	45
Lead time assumption	Dependent on capacity	43
	Deterministic	38
	Exponential	6
	General stochastic	9
	Not disclosed/Not applicable	17
	Normal	1
	Zero	14
Demand assumption	Erlang (Compound)	1
	Fuzzy	1
	General stochastic	32
	Poisson	79
	Poisson (compound)	9
	Stochastic	5
	Triangular	1
Number of customer classes	2	68
	3	2
	N	58
Customer response to stockouts	Backorders	53
	Backorders and lost sales	16
	Lost sales	57
	None	2
Service level constraint	None	104
	Waiting time	1
	α	8
	α and β	2
	β	11
	β and response time	2
Performance objective[1]	Cost	88
	Disease-adjusted life periods	1
	Profit	21
	Revenue	3
	Service levels	7
	Shortage	2
	Stock	9
Costs	Holding	106
	Shortage	100
	Ordering	35
Ordering policy type	Base-stock	20
	R, Q	4
	R, S	24
	s, Q	23
	s, S	3
	S-1, S	12
	State-dependent	27
	Dynamic base-stock	1
	Not applicable/Not disclosed	14
Ordering policy review	Continuous	84
	Periodic[4]	30
	Not applicable/Not disclosed	14
Optimal ordering policy?	No	90
	Yes	38
Multiperiod model?	No	18
	Yes	110
Rationing policy characteristics		
Type of policy	Dynamic	25
	State-dependent	45
	Static	58
Optimal rationing policy?	No	70
	Yes	58

[1] The summation does not total 128 because there are studies focusing on two (or more) categories.
[2] An exact method with an approximated (truncated) state-space.
[3] Refers to the activity of using numerical data to generate insights. Simulation is one method used to produce this data. See Section 3.3 for a detailed explanation.
[4] This total includes 28 studies using standard (R, S) or (R, Q) policies, plus two studies that derive an optimal state-dependent ordering policy within a periodic review framework.

model. Axsäter, Olsson, and Tydesjö (2007; 2004) explicitly considered a one-to-many 1-N network with one central warehouse and N customers located at different points, similar to Atan, Snyder, and Wilson (2018), who modelled a one-to-many 1-N network with transshipments; whereas other authors have focused on one-to-many production-inventory systems (Abouee-Mehrizi, Baron, and Berman 2014; Huang and Iravani 2007; Wang and Zhang 2021). In addition, van Wijk, Adan, and van Houtum (2013), Ayanso, Diaby, and Nair (2006), and Chen et al. (2011) studied a 2-N supply chain where one of the suppliers is a traditional central warehouse while the other is an emergency supply warehouse. Beyond these network structures, Shen, Yu, and Song (2022) investigated a distinct case of a serial inventory system with N stages and T periods, introducing inventory rationing policies for an N-stage multi-echelon problem.

3.1.2. Lead time assumptions

The way authors have modelled supply replenishment lead times depends on whether they considered capacity constraints or not. Capacitated models considered different phase-type probability distributions (Altiok 1985) to model the processing times of the capacitated server, such as Coxian, Erlang, exponential and mixed-Erlang, to keep the analysis tractable. However, most of the studies that did not consider capacitated supply processes assumed deterministic lead times (37 studies), 9 studies considered generally distributed stochastic lead times, 13 studies a lead time of zero, 6 studies considered exponentially distributed lead times, and one study considered a normally distributed lead time. Seventeen studies did not consider

any lead time as they were concerned only with a single period where no replenishments were modelled.

Most of the non-capacitated models have considered deterministic or zero lead times because of the added difficulty of modelling replenishment order crossovers (order arrivals have a different sequence than order placements) in scenarios with more than one outstanding replenishment order, e.g. when demand is significantly higher than Q considering an (s, Q) inventory policy or in scenarios with high demand variance and long inventory review periods, which is a possibility when facing stochastic lead times (Ayanso, Diaby, and Nair 2006). This is not generally an issue in most capacitated models as replenishment order arrivals are directly modelled by the queue.

3.1.3. Demand assumptions and number of customer classes

The majority of inventory rationing studies have modelled demand as individual Poisson arrivals (79 studies) because of its analytical tractability, e.g. using Markov decision processes, assuming that each customer arriving into the system requires only one unit of product. Besides these studies assuming individual Poisson arrivals, there are 9 studies that use a compound Poisson distribution to model the demand. Additionally, Hung and Hsiao (2013) model unitary arrivals with normally distributed order quantities. While 32 studies consider stochastic demand without making any specific distributional assumptions, others explicitly assume distributions such as triangular and uniform (Frank, Zhang, and Duenyas 2003), Erlang compound (ElHafsi, Camus, and Craye 2010), and even fuzzy (Cheng, Li, and Jiang 2016) demands. Additionally, three studies model stochastic demand by allowing at most one demand realisation per decision period, incorporating a probability of occurrence (Baron, Cire, and Savaser 2024; Wang and Yan 2009; You 2003). Furthermore, two studies consider demand as price-dependent (Elhafsi and Hamouda 2018; Liu et al. 2024), where price is a decision variable of the model.

Due to the difficulty of modelling problems with N number of customer classes, some studies (68) have only considered two customer classes, i.e. high- and low-priority, whereas 58 studies considered a finite number of N customer classes. The studies from Civelek, Karaesmen, and Scheller-Wolf (2015) and Lin et al. (2018) explicitly consider 3 classes of customers.

Furthermore, the majority of studies consider a system selling only one single type of product, an assumption that is common in inventory control models (Silver, Pyke, and Thomas 2016). However, eleven studies investigating production-inventory systems have extended this scope by incorporating demand for multiple product types. Some of these studies consider demand for components and finished products (Ceryan, Duenyas, and Koren 2012; Elhafsi et al. 2015; Elhafsi and Hamouda 2015; Kim and Kim 2022; Xu, Serrano, and Lin 2017), while others model demand for multiple finished products that share common components (Chen-Ritzo et al. 2011; ElHafsi, Camus, and Craye 2008; Fang and Elhafsi 2024; Nadar et al. 2018, 2014). Furthermore, the study by Lin et al. (2018) considers two end-products produced in two different facilities with different production rates. These studies typically assume that each component and finished product is produced using different facilities. In some cases, they further simplify the system by assuming zero assembly time for finished products (Chen-Ritzo et al. 2011; ElHafsi 2009; Elhafsi et al. 2015; Elhafsi and Hamouda 2015; ElHafsi, Camus, and Craye 2008; Fang and Elhafsi 2024; Nadar et al. 2018, 2014).

3.1.4. Customer response to stockouts

The customer stockout response models how customers react if there is no inventory on-hand to fulfil their requests. Customers can either elect to wait for a new replenishment of products and backorder their orders or abandon altogether their demand requirements, which translates into lost sales. In this regard, 53 studies considered models with backorders and 57 with lost sales. Moreover, 16 studies considered both backorders and lost sales models in the same study. Some of these studies investigated backorders and lost sales as separate cases (Benjaafar, ElHafsi, and Huang 2010; Fadıloğlu and Bulut 2010a; Topkis 1968; Vicil 2021a), while others modelled customer classes with distinct stockout responses, where certain customers were willing to wait while others were not (Carr and Duenyas 2000; Cattani and Souza 2002; Elhafsi and Hamouda 2018; ElHafsi, Fang, and Hamouda 2021; Enders et al. 2014; Kim and Kim 2022; Wang and Tang 2014; Xie et al. 2016). Additionally, some studies found that the structure of their optimal policies implied that demand for the same class(es) could either be backlogged or rejected depending on the system's current state (Ioannidis 2011; Ioannidis et al. 2021; Wang, Kulkarni, and Zhang 2023). Only the studies by Ayanso, Diaby, and Nair (2006) and Natarajan and Swaminathan (2017) did not explicitly model customer stockout response, as their problem settings assumed that unmet demand was always fulfilled through emergency shipments (e.g. drop-shipping or transshipments).

3.1.5. Service level constraints

Dealing with various types of customer classes with different priorities could create the need to offer differentiated service level agreements depending on the priority

of the customer class (Deshpande, Cohen, and Donohue 2003; Melchiors 2003). Because of this, some authors considered service level constraints in their models, looking to reach a minimum level of service to customers while still trying to minimise the total cost of the system under the inventory rationing policy. Studies dealing with inventory rationing policies have commonly considered two types of service levels (for further detail, see Lagodimos 1992):

- The α service level is the probability that a stockout does not occur at an arbitrary point in time (Escalona et al. 2021).
- The β service level, normally referred to as *fill rate*, is the percentage of demand that has been satisfied from available inventory.

The majority of inventory rationing studies did not consider service level constraints. The ones that did consider these constraints mostly concerned non-capacitated models: seven studies dealt with an α-service level constraint, whereas ten studies dealt with β-service level constraints and non-capacitated models, and two studies addressed both α and β service level constraints. The only study including a service level constraint regarding capacity-constrained models is the study by de Véricourt, Karaesmen, and Dallery (2001) who considered fill-rate constraints.

Gabor et al. (2018) and Vicil and Jackson (2018) introduce an additional service level constraint by incorporating a response time limit for low-priority customers while simultaneously enforcing a fill rate constraint for high-priority customers. In contrast, Kranenburg and van Houtum (2008) focus exclusively on a waiting time constraint.

3.1.6. Inventory ordering policies

Since inventory rationing decisions are highly interlinked with the replenishment policies as both have an influence on the performance of the inventory system, they have been evaluated together. For non-capacitated inventory models, most of the research on inventory rationing has assumed a specific replenishment policy and, under that assumption, has evaluated or found optimal rationing policies. Based on the classification provided by Silver, Pyke, and Thomas (2016), the most commonly assumed replenishment policy for non-capacitated models is the periodic review, order-up-to-level (R, S) policy, with the continuous review (s, Q) policy also being widely assumed. Out of the 83 studies that modelled a non-capacitated inventory system, only 3 set out to find an optimal replenishment policy. Assuming a periodic review replenishment policy, Goedhart, Haijema, and Akkerman (2022) and Abouee-Mehrizi et al. (2019) found that the optimal replenishment policy under periodic review was state-dependent based on inventory level. These two studies account for the periodic review models in our classification that do not fall under the standard (R,S) or (R,Q) policy types (see Appendix B for a detailed breakdown of policy combinations). For continuous review systems, Ding, Kouvelis, and Milner (2016) found that the optimal replenishment policy is a base-stock policy under a profit-maximising, continuous review, zero lead time, Poisson demand scenario and considering backorders.

A big proportion of studies (35 out of 45) concerned with capacitated models, on the other hand, have found that a (state-dependent) base-stock production policy is optimal (see, e.g. Ha 1997b). A notable exception regarding the studies investigating optimal replenishment policies is found in Pang, Shen, and Cheng (2014) who found that the optimal ordering policy is a reordering policy with fixed order size Q, under a profit-maximising, Poisson demand, Erlang production times, lost sales scenario.

Fourteen studies did not model a specific ordering policy, as they focused on single-period problems with exogenously defined initial inventory levels. In contrast, the remaining four studies on single-period problems assumed a periodic review (R, S) policy, which was also predetermined rather than derived within the model. For an overview of the number of studies with different inventory ordering and rationing policy combinations, refer to Appendix B.

3.1.7. Performance objectives and costs

The problem settings of inventory rationing studies have included various performance objectives. As shown in Table 1, cost has been the most commonly considered performance objective, followed by profit, whereas holding and shortage costs have been the most frequently considered type of costs. Although the main performance objective of several studies is not cost but service levels, some of these studies still accounted for different types of costs, e.g. Haynsworth and Price's (1989) objective was to reach a pre-specified service level but they still studied the shortage costs implications of their proposal.

In summary, the characterisation of inventory rationing problems shows a research stream focused on specific modelling choices, particularly the focus on non-capacitated systems, Poisson demand, and either full backordering or lost sales. These assumptions create the environment in which different rationing policies are designed and tested. The next section analyses the structure of these policies to understand how they are adapted to these different problem settings.

3.2. Classification and structure of rationing policies

The structure of rationing policies is dependent on problem characteristics and modelling assumptions, such as lead time assumptions, number of customer classes, and customer stockout response (Tan, Güllü, and Erkip 2009). This is particularly evident when comparing the structure of inventory rationing policies for non-capacitated systems against the rationing policies for capacitated (production-inventory) systems due to the fact that production policies need to also be considered in the latter. In this subsection, first, we review the characteristics of the policies for non-capacitated environments, followed by a revision of models with capacitated replenishment processes.

3.2.1. Non-capacitated inventory rationing models

Under a static policy, the critical levels remain fixed, regardless of the time elapsed since the previous replenishment. In contrast, dynamic policies adjust the critical level based on time, whereas state-dependent policies adjust the threshold based on other system variables, such as current inventory or backorder levels. It is noteworthy that Teunter and Klein Haneveld (2008) do not differentiate between dynamic and state-dependent policies in their classification, considering both to be forms of dynamic policies.

3.2.1.1. Dynamic inventory rationing policies. The critical level value for a dynamic policy depends on the elapsed time since the last replenishment period (or the time until the next replenishment arrival). Critical level values c_i thus gain an additional sub-index t compared to static policies, i.e. c_{it} is the critical level at time t in which all demand from customer classes below i is rejected. Table 2 presents an overview of the problems that have been addressed by using dynamic critical level policies and can be used as a reference to consult the assumptions regarding customer response to stockouts, service level constraints, number of customer classes, and demand and lead time probability distributions considered by the studies included in this review. It is worth noting that since some authors were concerned with single-period settings without modelling replenishments, the lead time distribution was not applicable/not relevant for some of those models, and thus these are shown as *NA* regarding lead time assumption in Table 2. From Table 2 it can also be seen that the studies from Alfieri, Pastore, and Zotteri (2017), Haynsworth and Price (1989), and Pibernik and Yadav (2009) are the only studies considering both a service level constraint and a dynamic critical level policy.

Topkis (1968) was the first study to find that a dynamic rationing policy is optimal in the single-period setting, a result that has been extended for different demand distribution settings and customer stockout responses in the single-period setting (see, e.g. Bao et al. 2018; Baron, Cire, and Savaser 2024; Teunter and Klein Haneveld 2008). Chew, Lee, and Liu (2013) also showed that a dynamic rationing policy is optimal for the multi-period, zero lead time setting with Poisson demands, N customer classes, and backorders.

Despite the optimality of dynamic rationing policies, modelling the cost, inventory level, and structure of the backorder functions for some environments could be analytically intractable or computationally intensive (Möllering and Thonemann 2008). Therefore, several authors developed simpler, non-optimal dynamic policies to cope with this complexity. For instance, Melchiors (2003) proposes a Restricted Time Remembering policy where the lead time period is divided into two periods instead of being modelled as m periods; whereas Fadıloğlu and Bulut (2010a) suggest that modelling the outstanding orders as if they are arriving continuously within the deterministic lead-time, i.e. the order is modelled as being received proportionally over the whole lead time period instead of being received in full only at the end of the lead time period, is equivalent to using a dynamic critical level.

Typically, a dynamic critical level threshold is a decreasing function over the time since the last replenishment, so less stock is reserved for higher priority customers the closer the system is to the next replenishment arrival (Alfieri, Pastore, and Zotteri 2017; Liu et al. 2015). However, Wang and Tang (2014) found contrasting results compared with this simple structure. They showed that when considering a mix of backorder and lost sales customers, and backorder costs depending on the backordered time, the critical level for customers with backorders could increase at some point in the replenishment cycle for certain customer classes. This is caused because a backlogged order at a particular instance in the replenishment period might incur a much higher penalty, and thus, it might be more convenient to reserve more stock for specific backorder-type classes.

Dynamic policies have mostly been applied to settings with deterministic or zero lead times and either periodic-review or single-period models because of the difficulty of modelling the structure of a dynamic critical level policy in a stochastic period. Thus, studies usually assume a deterministic and known review period in which to model the dynamic critical levels (see, e.g. Alfieri, Pastore, and Zotteri 2017; Chen, Xu, and Feng 2010; Hung et al. 2012; Liu et al. 2015). Four notable exceptions are the studies by Fadıloğlu and Bulut (2010a), Haynsworth

Table 2. Characterisation of studies on dynamic critical level policies in non-capacitated models.

Customer stockout response	Demand assumption	N	Deterministic lead time	Normal lead time	Zero lead time	NA (single period)
Backorders	General stochastic	*2*	Alfieri, Pastore, and Zotteri (2017)$^{\alpha}$	–	Chen, Xu, and Feng (2010)^{OPT*}	Kaplan (1969)^{OPT*}
		N	Hung et al. (2012)			Bao et al. (2018)^{OPT*}; Liu et al. (2015)
	Poisson	*2*	–	–	–	Liu and Zhang (2015); Teunter and Klein Haneveld (2008)^{OPT*};
		N	–	–	Chew, Lee, and Liu (2013)^{OPT*};	–
Lost sales	Stochastic	*2*	–	–	Liu et al. (2024)	–
		N	–	–	–	You (2003)^{OPT*}, Baron, Cire, and Savaser (2024)^{OPT*}
	General stochastic	*2*	–	–	Pibernik and Yadav (2009)$^{\alpha}$	Evans (1968)^{OPT*}
		N	–	–	–	Chen et al. (2012)^{OPT*}
	Poisson	*2*	Haynsworth and Price (1989)$^{\alpha}$	Hung and Hsiao (2013)	–	–
		N	Melchiors (2003); Wang, Tang, and Zhang (2015)	–	–	–
Backorders and lost sales	General stochastic	*N*	–	–	–	Topkis (1968)^{OPT*}
	Poisson	*2*	Fadıloğlu and Bulut (2010a)	–	–	–
		N	–	–	Wang and Tang (2014); Wang, Kulkarni, and Zhang (2023)^{OPT*}	–
None	General stochastic	*2*	–	–	Natarajan and Swaminathan (2017)	–

OPT*: Study deriving an optimal rationing policy
α: Study considering a service level constraint α

and Price (1989), Melchiors (2003), and Wang, Tang, and Zhang (2015) who investigate the performance of dynamic critical level policies in a continuous review setting by characterising the dynamic critical level policy only inside the period of the deterministic replenishment lead time.

In Table 2, 'Stochastic' demand refers to cases where the demand process does not follow a standard probability distribution. For instance, You (2003) modelled demand as a single potential arrival per period, occurring with a certain probability, while also allowing lower-priority customers to upgrade to a higher class with some probability by paying a premium. Similarly, Liu et al. (2024) classified demand as 'Stochastic' by modelling class-specific demand arrivals as price-dependent, where price itself is a dynamic decision variable alongside replenishment and inventory policies.

Overall, the literature shows that dynamic policies are typically found to be optimal only in settings with deterministic or zero lead times, or in single period settings, raising the question of how the optimal policy structure changes when lead time is uncertain.

3.2.1.2. State-dependent inventory rationing policies.

In non-capacitated settings, the structure of optimal rationing policies is highly dependent on the presence of lead times. While dynamic rationing policies have been shown to be optimal for certain single-period problems or settings with zero lead time, the introduction of a non-zero lead time (deterministic or exponentially distributed) fundamentally alters the optimal policy structure. Specifically, when lead times are considered, the optimal policy becomes state-dependent, meaning that inventory levels and outstanding backorders, in the backorders setting, directly influence rationing decisions. Thus, in this study, we make a distinction between dynamic policies (those dependent only on time) and state-dependent policies, which are explicitly modelled as dependent on other state variables such as inventory and backorder levels, contrasting with previous research that regarded both as dynamic policies (Teunter and Klein Haneveld 2008). Table 3 provides an overview of the studies that have considered/derived state-dependent policies for non-capacitated models.

State-dependent rationing policies have also been found to be optimal in special single-period and zero lead

Table 3. Characterisation of studies on state-dependent policies in non-capacitated models.

Customer stockout response	Demand assumption	N	Deterministic lead time	Exponential lead time	Zero lead time	NA (single period)
Backorders	General stochastic	*N*	Abouee-Mehrizi et al. (2019)^{OPT}; Shen et al. (2019)^{OPT*}	–	Shen, Yu, and Song (2022)^{OPT*}	–
	Poisson	*N*	–	–	Ding, Kouvelis, and Milner (2016)^{OPT}	–
Lost sales	Stochastic	*2*	Wang and Yan (2009)^{OPT*}	–	–	–
	Poisson	*2*	Goedhart, Haijema, and Akkerman (2022)^{OPT}	–	–	Chen et al. (2011)^{OPT*}
		N	–	van Wijk, Adan, and van Houtum (2013)^{OPT*}	–	–
	Poisson (compound)	*2*	–	Kouki and Larsen (Kouki and Larsen 2021)$^{\beta}$	–	–
	Triangular	*2*	–	–	Frank, Zhang, and Duenyas (2003)^{OPT*}	–
Backorders and lost sales	General stochastic	*2*	–	–	Xie et al. (2016)^{OPT*}	–

OPT: Study deriving optimal replenishment and rationing policies
OPT*: Study deriving an optimal rationing policy
β: Study considering a fill rate constraint β

times settings. For example, Chen et al. (2011) analyse a system with two retailers that can drop-ship inventory to an e-tailer and determine the optimal rationing policies for each in a revenue-maximising, single-period setting. They find that the optimal policy for the first (preferred) retailer is a dynamic rationing policy. However, for the second (non-preferred) retailer, the optimal policy is state-dependent on the inventory levels of the first retailer and the remaining periods. You (2003) also examines a revenue-maximising problem but since You (2003) considers a setting where customer classes can upgrade their own class by paying a premium and assumes a single inventory pool for all customer classes, they find that the optimal policy remains purely dynamic.

In zero-lead-time settings, the distinction between when a state-dependent policy or a dynamic policy is optimal depends on whether the customer classes exhibit heterogeneous responses to stockouts. Studies by Ding, Kouvelis, and Milner (2016), Shen, Yu, and Song (2022), and Xie et al. (2016) account for differences in customer stockout response, such as higher impatience levels among certain customer classes, which leads to the optimality of inventory- and backorder-dependent rationing policies. Similarly, Frank, Zhang, and Duenyas (2003) found a state-dependent critical level policy to be optimal in a zero-lead-time setting where all deterministic high-priority demand must be fulfilled, while stochastic low-priority demand is subject to lost sales. In contrast, studies that assume homogeneous stockout responses across all customer classes in a zero-lead-time setting have found a dynamic rationing policy to be optimal (Chen, Xu, and Feng 2010; Chew, Lee, and Liu 2013; Wang, Kulkarni, and Zhang 2023).

Along with identifying the optimality of state-dependent rationing policies, three studies have also derived optimal ordering policies under specific problem settings. In contrast, research focused on deriving dynamic rationing policies has primarily assumed predefined ordering policies rather than deriving them as part of the optimal solution framework. Thus, Table 3 highlights a distinction between the studies that derived the optimal ordering policy along the optimal rationing policy (shown as OPT in Table 3) and the studies that assumed an ordering/replenishment policy to derive an optimal *rationing* policy (shown as OPT∗ in Table 3). In this regard, Ding, Kouvelis, and Milner (2016) found that a base-stock ordering policy is optimal for a continuous review, backordering, and profit-maximising setting with zero lead times; Abouee-Mehrizi et al. (2019) show that the optimal ordering policy is a state-dependent policy with order sizes dependent on inventory levels for a setting periodic review replenishment, backordering, cost-minimising setting with deterministic lead times; whereas Goedhart, Haijema, and Akkerman (2022), assuming an ordering period with 7 time periods, a periodic review policy, lost sales, profit-maximising setting with deterministic lead times shows that the optimal ordering policy is state-dependent on inventory levels. They suggest that the optimal ordering policy resembles

an (R, Q) policy with low levels of inventory and a base-stock policy with high inventory levels. The complexity of deriving such optimal state-dependent policies, however, has led many researchers to investigate simpler, more practical heuristics.

3.2.1.3. Static inventory rationing policies. The fact that dynamic and state-dependent critical level policies are extremely difficult to model for some settings (Fadıloğlu and Bulut 2010a; Möllering and Thonemann 2008) and that they could be difficult to implement in real environments (Melchiors 2003), has motivated some authors to propose static policies, which are simpler to both estimate and operate. Table 4 provides an overview of the problems under which static rationing policies have been proposed.

While these policies are not optimal, they are much easier to implement than dynamic policies. This implies that they are more applicable in practical settings as they have been shown to significantly improve the performance of inventory systems with multiple customer classes with differing shortage penalties when compared with systems with no inventory rationing (Dekker et al. 2002; Möllering and Thonemann 2008). Furthermore, static policies have also been shown to be very close in terms of performance with the optimal dynamic policy in some settings. For instance, Frank, Zhang, and Duenyas (2003) found a 3% maximum difference between the optimal dynamic policy and a static critical level policy while considering moderate ordering costs; whereas Enders et al. (2014) found an average difference of 2.09% between the optimal and the static rationing policies while reaching optimal costs in 45% of the cases that static policies were used.

An interesting case of a static policy comes from Ghosh, Piplani, and Viswanathan (2015) where, instead of considering a single stock pool for the high- and low-priority class, they propose a model with 'two bins'. The first bin is exclusively assigned to fulfil the demand of the high-priority class. As long as the first bin has stock, the second bin only fulfils the demand of the low-priority class; otherwise, the second bin fulfils the demand of both classes under a static critical level policy. While this policy performed significantly better than the single critical level policy in terms of low-priority fill rate and slightly better in terms of high-priority fill rate, it suffered from higher total costs.

The lower complexity of modelling static rationing policies also facilitates their use in assessing differentiated service level constraints as 19 out of the 23 studies that considered service level constraints in non-capacitated systems adopted static rationing policies.

3.2.2. Capacitated inventory rationing models

In contrast to the non-capacitated models discussed previously, a distinctive feature of capacitated production-inventory models is that decisions about when and whether to produce are integrated with the rationing policy. Consequently, most studies on production-inventory systems have investigated/derived state-dependent policies, since optimal production control and inventory allocation rules often depend on system states (e.g. on-hand inventory, backorders, state of machines). In contrast, simpler static or dynamic threshold policies have been explored only occasionally, despite evidence that they can perform well compared to simpler policies. For instance, de Véricourt, Karaesmen, and Dallery (2001) show that even non-optimal multilevel policies can outperform FIFO and strict priority rules in a backorders setting. Table 5 presents a summary of the models considering capacitated systems and static or dynamic rationing policies. Since all of the studies included in Table 5 consider Poisson arrivals, the demand distribution is not shown.

Static policies have also been shown to be optimal in production-inventory systems for some specific cases. For instance, Ha (1997b) shows that a static policy is optimal in an environment with one server with exponential processing times (e.g. one production facility), N customer classes with Poisson arrivals requesting one unit of product, and lost sales; whereas for the same problem characterisation but with backorders, some authors show that a switching-curve-type critical level policy dependent (and decreasing) on the number of backorders of each customer class is the optimal policy (de Véricourt, Karaesmen, and Dallery 2002; Ha 1997a).

The structure and properties of switching-curve policies for two customer classes in the backorders setting tend to be straightforward to implement and somewhat easy to understand (although not easy to develop). For instance, the production-rationing policy derived by Ha (1997a) suggests the following: '*It is always optimal to produce for class 1 backorders. When there are only class 2 backorders, it is optimal to produce for more inventory if the inventory level is below S(y)*' – the switching curve depending on y class 2 backorders – '*and to produce to satisfy class 2 backorders otherwise. When there is no backorder, it is optimal to produce for inventory if the inventory level is below S(0)*' –the base-stock level– '*and to stop otherwise. It is optimal to satisfy an incoming class 2 customer from on-hand inventory if the inventory level is above* $S(y+1)$ *and to backorder for this customer otherwise*'.

With more than two customer classes (or components), however, the optimal policy typically involves multiple nested thresholds (or switching) levels with

Table 4. Characterisation of studies on static critical level policies in non-capacitated models.

Customer stockout response	Demand assumption	N	Deterministic lead time	Exponential lead time	General stochastic lead time	Zero lead time	NA (single period)
Backorders	General stochastic	*2*	Escalona et al. (2019)$^{\alpha,\beta}$; Escalona, Ordóñez, and Iturrieta (2017a)$^{\alpha}$; Escalona, Ordóñez, and Kauak (2017b); Möllering and Thonemann (2010$^{\beta}$, 2008); Tempelmeier (2006)$^{\beta}$; Wang et al. (2013b)$^{\alpha}$; Wang et al. (2020)	–	–	–	–
		N	–	–	Moon and Kang (1998)	–	–
	Poisson	*2*	Dekker, Kleijn, and de Rooij (1998); Deshpande, Cohen, and Donohue (2003); Fadıloğlu and Bulut (2010b); Fadıloğlu and Bulut (2019); Gabor et al. (2018)$^{\beta}$; Ghosh, Piplani, and Viswanathan (2015)$^{\beta}$; Nahmias and Demmy (1981); Vicil (2021b)$^{\beta}$; Wan and Cao (2018); Wang et al. (2013a)$^{\alpha}$	Dendauw et al. (2021); Vicil and Jackson (2016)$^{\beta}$	Vicil and Jackson (2018)$^{\beta}$; Vicil (2021c)	–	–
		N	Arslan, Graves, and Roemer (2007)$^{\beta}$; Axsäter, Kleijn, and de Kok (2004); Mishra and Sahab (2011)	–	–	–	–
	Poisson (compound)	*2*	Axsäter, Olsson, and Tydesjö (2007), Berling, Johansson, and Marklund (2023)$^{\beta}$	–	–	–	–
Lost sales	Fuzzy	*2*	Cheng, Li, and Jiang (2016)$^{\alpha}$	–	–	–	–
	General stochastic	*2*	Tan, Güllü, and Erkip (2009)	–	–	–	Ma and Jemai (2019)
		3	–	–	–	Civelek, Karaesmen, and Scheller-Wolf (2015)	–
		N	–	–	–	–	Chen-Ritzo et al. (2011)
	Poisson	*2*	Atan, Snyder, and Wilson (2018); Melchiors, Dekker, and Kleijn (2000)	Isotupa (2015); Liu and Wei (2019)	–	–	Samii et al. (2012); Samii (2016)
		N	–	–	Dekker et al. (2002)$^{\beta}$; Kranenburg and van Houtum (2008, 2007^{W})	–	Schulte and Pibernik (2017, 2016$^{\alpha}$)$^{\beta}$
Backorders and lost sales	General stochastic	*N*	–	–	–	Veinott (1965)	–
	Poisson	*2*	–	–	Enders et al. (2014); Vicil (2021a)$^{\beta}$	–	–
None	General stochastic	*2*	–	–	Ayanso, Diaby, and Nair (2006)	–	–

α: Study considering a service level constraint α
β: Study considering a fill rate constraint β
W: Study considering a waiting/response time constraint

Table 5. Characterisation of studies on static and dynamic critical level policies in capacitated models.

Type of policy	Processing time distribution	N	Backorders	Lost sales	Backorders and lost sales
Static	Coxian	*N*	–	Lee and Hong (2003)	–
	Exponential	*2*	–	Ahmadi and Shavandi (2014)^{OPT}	Cattani and Souza (2002); Ioannidis (2011); Ioannidis et al. (2021)
		N	de Véricourt, Karaesmen, and Dallery (2001)$^{\beta}$; Sahba, Balcıoğlu, and Banjevic (2018)	Ha (1997b)^{OPT}; Turgay, Karaesmen, and Örmeci (2015)^{OPT}	–
	General stochastic	*N*	Abouee-Mehrizi, Baron, and Berman (2014)	–	–
Dynamic	Mixed-Erlang	*N*	–	Pang, Shen, and Cheng (2014)^{OPT}	–

OPT: Study deriving optimal replenishment and rationing policies
β: Study considering a fill rate constraint

as many as *N−1* distinct threshold levels (out of *N* number of customer classes), making it conceptually and computationally more complex to derive, visualise, and implement. In this regard, the underlying Markov decision process can require tracking multiple backlog dimensions (and inventories in the case of settings with component commonality in assemble-to-order problems), so the policy's switching criteria may not be as straightforward as in the case of two classes.

To provide a structured overview of the findings on state-dependent rationing policies, Table 6 summarises the main characteristics of studies that have derived state-dependent rationing policies for capacitated systems. Research focusing on production-inventory systems has gone beyond the study of a single-server, single-product, unitary demand problem introduced by Ha (1997a), for the backorders case, or by Ha (1997b), for the lost sales case. For instance, for the lost sales case, Huang and Iravani (2007) consider a two-echelon system of one production site and two retailers ordering under an (s, Q) policy, which introduces the issue of rationing quantity, as the customers' orders are not unitary. For this problem, an inventory level-dependent rationing policy was found to be optimal, in contrast to the optimal static policy in Ha (1997b). Other extensions to the work by Ha (1997a; 1997b) for both the lost sales and backorders case with Poisson demands can be found in Table 6 under the categories of Poisson demand, exponentially-distributed processing times, and inventory (backorder)-dependent rationing policies.

Table 6 also shows that state-dependent policies can depend on other variables in addition to inventory and backorder levels due to the configuration of the production system. For instance, production and rationing policies may depend on the inventory levels and state of production stages for Erlang-distributed processing times (Gayon, de Véricourt, and Karaesmen 2009; Ha 2000), on whether a machine is available when considering unreliable machines (Cheng, Gao, and Shen 2011a), on the current stage of demand in a setting where information is shared about the evolution of Erlang-compound demand downstream (ElHafsi, Camus, and Craye 2010), or on the number of active servers in a production system with N parallel production sites (Bulut and Fadiloğlu 2011).

Other studies consider more complex assemble-to-order production-inventory systems where the rationing policies depend on the inventory state of the common components needed to assemble the end product, under the assumption that the final assembly of components is instantaneous (but not the production of the components) (see, e.g. Benjaafar and ElHafsi 2006; ElHafsi 2009), where some of the components might also face stochastic demand (see, e.g. Elhafsi et al. 2015; Elhafsi and Hamouda 2015; Xu, Serrano, and Lin 2017), or where there is demand for more than one end product with component commonality (see, e.g. ElHafsi, Camus, and Craye 2008; Nadar et al. 2018). A particular case is studied by Kim and Kim (2022) who study a system where there is only one end product and one customer class for the end product but the semi-finished product has stochastic demand, which creates state-dependent rationing levels to reserve semi-finished inventory to complete the production of end products.

In addition, Benjaafar et al. (2011) and Ceryan, Duenyas, and Koren (2012) overcome the assumption of an instantaneous assembly task in an assemble-to-order production system producing only one end product with multiple components by considering exponentially-distributed processing times for the final assembly and component production. This is why we classified the processing time distribution of this study as 'Stochastic' in Table 6, as it does not follow a common probability distribution due to the probabilistic time of matching of all components. Another special case can be found in the study by Elhafsi and Hamouda (2018), for which

Table 6. Characterisation of studies on state-dependent critical level policies in capacitated models.

Demand assumption	Processing time distribution	State description	Classes	Backorders	Lost sales	Backorders and lost sales
Erlang-distributed inter-arrival times	Erlang	Inventory and stage of demand	*N*	–	ElHafsi, Camus, and Craye (2010)^{OPT}	–
General stochastic	Zero	Inventory	*N*	–	Zhou, Lee, and Wu (2011)^{OPT*}	–
	Deterministic	Inventory and capacity levels	*N*	–	Gao, Xu, and Ball (2012)^{OPT*}	–
Poisson	Exponential	Inventory	*2*	–	Huang and Iravani (2007)^{OPT}; Wang and Zhang (2021)^{OPT}; Yu et al. (2017)^{OPT}	Carr and Duenyas (2000)^{OPT}
			N	–	Gayon, Benjaafar, and de Véricourt (2008)^{OPT}	–
		Inventory and backorders	*2*	Ha (1997a)^{OPT}	–	Benjaafar, ElHafsi, and Huang (2010)^{OPT}
			N	de Véricourt, Karaesmen, and Dallery (2002)^{OPT}	–	ElHafsi, Fang, and Hamouda (2021)^{OPT}
		Inventory of the common component and finished products, backorders for low-priority product	*2*	Fang and Elhafsi (2024)^{OPT}	–	–
		Inventory and backorders of components and finished products	*N*	Elhafsi and Hamouda (2015)^{OPT}	–	–
		Inventory and stage of production	*N*	–	Xu, Serrano, and Lin (2017)^{OPT}	–
		Inventory of components	*N*	–	Benjaafar and ElHafsi (2006)^{OPT}; Elhafsi et al. (2015; 2008)^{OPT}; Nadar, Akan, and Scheller-Wolf (2014)^{OPT}	–
		Inventory, backorders, and capacity levels	*N*	Mayorga and Ahn (2011)^{OPT}	–	–
		Inventory, number of active servers	*N*	–	Bulut and Fadiloğlu (2011)^{OPT}	–
	Exponentially distributed TTF and TTR	Inventory (finished products) and state of machines	*N*	–	Cheng, Gao, and Shen (2011a)^{OPT}	–
		Inventory of components and state of machines	*N*	–	Cheng, Gao, and Shen (2011b)^{OPT}	–
	Erlang	Inventory, backorders and stage of production	*N*	Gayon, de Véricourt, and Karaesmen (2009)^{OPT}	–	–
		Inventory and stage of production	*N*	–	Ha (2000)^{OPT}	–
		Inventory of components	*N*	–	Nadar et al. (2018)	–
	Stochastic	Inventory of components and finished products	*N*	–	–	Ceryan, Duenyas, and Koren (2012)^{OPT}
Poisson (compound)	Exponential	Inventory	*2*	Huang and Iravani (2008)^{OPT}	–	–
			3	–	Lin et al. (2018)^{OPT}	–
		Inventory and backorders	*N*	Xu et al. (2010)^{OPT}	–	–
		Inventory of components	*N*	–	ElHafsi (2009)^{OPT}	–
		Inventory of semi-finished product, backorders of semi-finished and end products	*1*	–	–	Kim and Kim (2022)^{OPT}
	Stochastic	Inventory of components and finished products	*N*	–	Benjaafar et al. (2011)^{OPT}	–
Stochastic	Exponential	Inventory and backorders	*2*	–	–	Elhafsi and Hamouda (2018)^{OPT}

the demand has been classified as 'Stochastic' because they consider two different types of demand: a long-term demand, with Erlang-distributed inter-arrival times, and a walk-in demand with Poisson arrivals with a price-dependent rate, being that the price also depends on the state of the system.

As Tables 5 and 6 show, most of the studies that considered capacitated models assumed exponentially distributed processing times to keep the analysis tractable. Overcoming this limitation, a number of studies considered phase-type distributions to model processing times in capacitated systems, e.g. Coxian, Erlang, and Mixed-Erlang. Interestingly, Ha (2000) found that rationing levels exhibit low sensitivity to variations in the shape parameter of the Erlang distribution, suggesting that Erlang-based approximations, such as those proposed by Ha (2000) and Gayon, de Véricourt, and Karaesmen (2009) provide a tractable means for evaluating rationing performance in systems with generally distributed processing times (e.g. the M/G/1 case). While the shape parameter influences cost due to its effect on variability (higher variability tends to increase costs) these findings indicate that optimal rationing levels remain relatively stable across different distributional shapes.

The numerical experiments conducted by Ha (2000) and Gayon, de Véricourt, and Karaesmen (2009) demonstrate that even simple, static critical-level policies with optimised parameters perform remarkably well, with at most a 2% increase in average costs compared to the optimal state-dependent policy. Moreover, Gayon, de Véricourt, and Karaesmen (2009) found that the FIFO policy performs significantly worse than the optimal inventory rationing policy, with a performance degradation of up to 23% in scenarios with three customer classes. These findings for capacitated systems align with insights from non-capacitated models: implementing an inventory rationing policy, regardless of whether it is static or dynamic, offers a performance advantage over FIFO.

3.3. Methodological approaches

Based on the taxonomy of de Kok et al. (2018), four modelling methods for studying inventory rationing problems can be identified, namely, exact analysis, approximations, computer simulations, and numerical experiments. A study was classified as using an exact method if the objective function could be evaluated exactly. Exact methods were further categorised into Mathematical/Algebraic Analysis, Dynamic Programming, Markov Decision Processes, Markov Chain Analysis, and Mathematical Programming. A large proportion of the studies included in this review used either Markov Decision Processes (37) or Markov Chain Analysis (24) to derive or evaluate inventory rationing policies. These methods are especially well suited to the stochastic, event-driven structure of arrivals and replenishments, particularly when lead times follow phase-type distributions and demand arises from (compound) Poisson processes.

Markov chain-based approaches involve two main considerations. First, in continuous-time models, some studies discretise time into sufficiently small intervals to ensure that the probability of multiple arrivals within a single interval remains negligible (see, e.g. Bulut and Fadiloğlu 2011; Wang and Zhang 2021). Although there is always a nonzero probability of more than one arrival occurring in a period, we did not classify this discretization as an approximation. Second, when backorders are allowed, the state space can become infinitely large, needing truncation of the state space either in the model formulation or the numerical experiments (referred to as 'Approximate exact model (truncation)' in Table 1) (see, e.g. Möllering and Thonemann 2010; Pang, Shen, and Cheng 2014).

On the other hand, we classify a method as approximate when the objective (or service levels) cannot be evaluated exactly or becomes too cumbersome to calculate, thus relying on approximate expressions (e.g. approximate closed-form formulas for service levels or steady-state probabilities). Note that a non-optimal heuristic designed to mirror the behaviour of an optimal policy can still yield an exact cost calculation if the system's structure is well understood. In that case, since the resulting performance measure can be computed precisely, it was labelled as an exact method. For instance, Abouee-Mehrizi, Baron, and Berman (2014) analyse the cost in an exact manner using the Queueing Decomposition method (Abouee-Mehrizi and Baron 2016) because they mention that dynamic programming is 'less practical' when considering the generally distributed service times for M/G/1 queues, whereas they find the optimal parameters of the inventory policies in an enumerative manner.

We define numerical experiments as the activity of using quantitative data, whether from analytical models, approximations, or simulations, to explore a model's behaviour or compare policy performance under various scenarios. In contrast, simulation is the specific method used to generate results without direct analytical evaluation. Thus, several of the 95 studies that conduct numerical experiments use simulation as the tool to generate the performance data. Simulation is commonly used alongside approximate methods, particularly when the system cannot be evaluated exactly due to its complexity or when closed-form expressions are unavailable. The primary uses of simulation in the reviewed studies include

sensitivity analysis, validating the accuracy of approximations by estimating system performance, computing objective values in analytically intractable settings or estimating steady-state probabilities when direct analytical derivation is impractical.

Despite the fact that 95 studies carried out numerical experiments to test the quality of their solutions, only eight studies (see: Alfieri, Pastore, and Zotteri 2017; Ayanso, Diaby, and Nair 2006; Baron, Cire, and Savaser 2024; Chen-Ritzo et al. 2011; Dekker, Kleijn, and de Rooij 1998; Kaplan 1969; Kranenburg and van Houtum 2008; Wang et al. 2020) explicitly considered empirical data for their experimentation, whereas the rest used synthetic data. However, some authors were inspired by different realistic problems, e.g. spare parts (Dekker et al. 2002, 1998; van Wijk, Adan, and van Houtum 2013), omni-channel retail (Berling, Johansson, and Marklund 2023; Goedhart, Haijema, and Akkerman 2022), blood donation (Civelek, Karaesmen, and Scheller-Wolf 2015), or vaccines (Samii et al. 2012).

4. Discussion

This section discusses the main findings of this study by summarising the answers to our research questions as well as future research directions and the study's implications. The review shows a field that is continuously developing, with recent studies tending to extend existing problem settings to more complex and modern operational contexts, such as omni-channel fulfilment, dynamic pricing environments, and assemble-to-order systems.

In response to RQ1, our analysis shows that the main problem characteristics are largely defined by the distinction between capacitated and non-capacitated environments and a focus on either backordering or lost sales for customer stockouts. The majority of studies in this review focus on non-capacitated environments, reflecting a broader trend in inventory management research (de Kok et al. 2018). Within non-capacitated systems, most studies (62 out of 83) evaluate the performance of rationing policies rather than deriving an optimal policy, though 21 studies do present optimal policies. Non-capacitated models tend to focus more on backordering scenarios, with 42 studies examining backorders compared to 33 focusing on lost sales. This result may be linked to the fact that many non-capacitated studies explore non-optimal policies, particularly static critical level policies. Among the 42 non-capacitated studies focusing on backorders, 29 employ static critical level policies, likely due to the difficulty of optimally tracking outstanding backorders in a state-dependent manner. In such cases, static policies serve as a more practical alternative, offering ease of implementation and analysis while still providing significant performance benefits over a non-rationing baseline, as shown by studies considering non-capacitated (Enders et al. 2014; Frank, Zhang, and Duenyas 2003) and capacitated settings (Gayon, de Véricourt, and Karaesmen 2009; Ha 2000). In the case of capacitated models, it can be seen that the majority of studies that considered a production-inventory system also found optimal policies (37 out of 45). These results were obtained mainly by considering some simplifying assumptions such as (compound) Poisson demands and exponential processing times to make the analysis tractable using Markovian analysis.

Regarding RQ2, our review classifies inventory rationing policies as static, dynamic, or state-dependent, with the optimal structure being highly sensitive to the presence of lead times. An interesting insight emerging from the literature is that dynamic policies have been found to be optimal in specific, non-capacitated cases, such as single-period problems or zero lead time settings. However, introducing non-zero lead times, whether deterministic or exponentially distributed, significantly alters the optimal rationing structure to a state-dependent policy. On the other hand, a number of studies considering the continuous review (s, Q) ordering policy with deterministic lead times under a non-capacitated system (Atan, Snyder, and Wilson 2018; Melchiors 2003; Melchiors, Dekker, and Kleijn 2000; Wang, Tang, and Zhang 2015) simplify their analysis by assuming that at most one order can be outstanding at any given time.

These observations highlight a trade-off in the modelling approaches used across the literature. Studies aiming to derive optimal policies often impose simplifying assumptions. Conversely, studies prioritising practical policy evaluation tend to explore more general settings, incorporating additional complexities such as service level constraints, at the expense of optimality guarantees.

Studies that extend beyond standard simplifying assumptions have provided valuable insights into the sensitivity of inventory models to various distributional attributes. For instance, research incorporating compound Poisson demand processes has examined the influence of order sizes on inventory rationing policies. In production-inventory systems, ElHafsi (2009) investigated the effects of order batch sizes and the coefficient of variation in order sizes on the optimal average cost. Their findings show that increases in both factors lead to higher total average costs, and notably, the cost difference between a FIFO policy and the optimal policy narrows as order size variability increases. This is particularly interesting because higher variability typically forces an increase in base-stock levels, which in turn raises holding costs, thereby reducing the benefits of reserving stock

for high-priority customers. Similarly, Huang and Iravani (2008) showed that both base-stock and rationing levels rise when fixed order sizes for high-priority customers increase. In addition, ElHafsi, Camus, and Craye (2010) concluded that reducing the variability of order inter-arrival times and processing times was much more influential to cost reduction than the variability of order sizes.

In non-capacitated systems, the impact of lead time variability on decision parameters has also been explored. Enders et al. (2014) found that optimal base-stock levels are more sensitive to increases in lead time variability than optimal rationing levels when lead times are modelled using a degenerate hyperexponential distribution. However, rationing levels remain somewhat affected. To illustrate this effect, increasing the lead time's squared coefficient of variation from 1 (an exponential distribution) to 10 changed the optimal rationing level in approximately 20% of the scenarios studied. Additionally, Ayanso, Diaby, and Nair (2006) used simulation to investigate the joint effects of demand and lead time variability on threshold levels in a profit-maximising system without backorders or lost sales (with drop-shipping allowed to satisfy demand). They found that lead time variability has a more substantial influence on threshold levels than does high-priority demand variability, with higher variabilities yielding higher thresholds and an interaction between the two factors further amplifying this effect.

These studies show that moving beyond conventional assumptions can influence the operational parameters of inventory rationing systems as well as the performance advantages of optimal (and suboptimal) rationing over policies such as FIFO. In fact, multiple studies have quantified significant improvements of rationing over FIFO: some have demonstrated that the optimal policies can drastically reduce costs relative to FIFO (see, e.g. Benjaafar and ElHafsi 2006; Ha 1997b; van Wijk, Adan, and van Houtum 2013), while others have shown that even relatively simple, non-optimal rationing rules yield considerable benefits when compared to a FIFO approach (see, e.g. Cattani and Souza 2002; Enders et al. 2014; Wan and Cao 2018).

A notable exception to this pattern is found in Xu, Serrano, and Lin (2017), who studied a tandem production system in which the intermediate and end products face independent demand with lost sales, and each product type has distinct customer classes. Their numerical experiments reveal that, in this particular setting, the FIFO policy outperforms a static critical level policy applied separately to each product type, where each product has a distinct set of rationing levels corresponding to its customer classes. This result highlights the importance of studying inventory rationing policies under alternative system configurations and assumptions to better understand their implications. In the next section, we outline several avenues for future research, addressing gaps created by common modelling assumptions and exploring new directions to further advance this field.

4.1. Future research opportunities

In response to RQ3, this section details several promising avenues for future research that emerge from our review. One opportunity arises from the common simplifying assumptions employed in inventory rationing studies, which may limit the applicability of current findings. As discussed earlier, unitary Poisson demand arrivals and exponential processing times have been widely used to maintain analytical tractability, even when approximations are involved or when the focus is on performance evaluation rather than deriving an optimal policy. However, for many settings, unitary Poisson demands (Cattani, Jacobs, and Schoenfelder 2011) and exponential processing times (Inman 1999) may not accurately reflect reality. Given that geometric-Poisson demands have been shown to better represent realistic demand patterns (Lengu, Syntetos, and Babai 2014; Prak et al. 2021) and Erlang-distributed processing times could be more realistic (Gayon, de Véricourt, and Karaesmen 2009), more research is needed to explicitly investigate how various demand and processing time (or lead time) distributional attributes affect ordering, production, and rationing decisions, thereby extending previous studies conducted under simplified scenarios.

Moreover, there is a notable gap in research on capacitated systems with service level constraints, even though research focusing on this topic has shown very interesting conclusions (see, e.g. de Véricourt, Karaesmen, and Dallery (2001), who demonstrated that under a backorder scenario, FIFO is equivalent to a static rationing policy when both customer classes have a fill rate of 80%). Furthermore, since calculating relative differences in terms of backorder costs between customer classes could be challenging (de Véricourt, Karaesmen, and Dallery 2001), future research could investigate the equivalence between considering shortage costs and fill rates, as suggested by Escalona et al. (2017a). Given the practical importance of service level agreements (Chen and Thomas 2018) and service differentiation, this topic presents a promising opportunity for future research in capacitated settings.

A related but largely unexplored area of research concerns the impact of current lost sales caused by the selected inventory policy on future demand (see, e.g. Sokolinskiy et al. 2019), particularly the erosion of customer goodwill among lower-priority segments when

there is a significant disparity between shortage penalties or fill rate targets under rationing policies. In such contexts, incorporating dynamic pricing strategies (see, e.g. Liu et al. 2024; Xie et al. 2016) may be critical to attracting lower-priority customers who might otherwise be lost due to poor service levels.

Inventory rationing studies considered single-product models just like the traditional models in inventory management (Jalali and Van Nieuwenhuyse 2015; Silver, Pyke, and Thomas 2016). For inventory control policies, this is logical since these policies can be applied independently to control the stock of each SKU sold by a firm. However, this assumption could be flawed when one supplier replenishes multiple products to the same network using the same production capacity, since the supplier will need to allocate production to both the customer class (or retailer) and the product type. Consequently, more research is needed on multi-product (or multi-component) systems with shared capacity, beyond the studies where end products or components are produced in individual facilities, which have been included in this review, to better understand the impact of shared resources on production and rationing policies.

Furthermore, complexities in cost structures and customer behaviour often present additional challenges. For example, future research could investigate settings where the penalty for not serving high-priority customers is substantially greater (e.g. sedatives for high-risk patients in hospitals) and where lower-priority customers have access to product substitutes. In such cases, although the critical (or rationing) levels may be much higher, some low-priority demand might still be fulfilled by substitutes even if the main product remains the preferred option. This interplay between severe penalties and product substitutes has not been sufficiently investigated. In addition, certain environments involve customers who incur increasing penalties the longer their backorders persist, e.g. in healthcare settings where delays in receiving critical products can lead to further deterioration in a patient's health. For these cases, additional research is needed to understand how incremental shortage costs, which increase over the duration of backorders, influence the optimal parameters of inventory policies for customer classes with different priorities.

Similarly, the interplay between perishability and critical level thresholds has yet to be explored. Reserving stock for high priority customers might be less valuable the closer the decision epoch is to the expiration date due to the increased probability of spoilage. This interplay introduces a dynamic tension between prioritising service levels and minimising waste, suggesting that optimal rationing thresholds may need to adapt not only to customer class but also to remaining shelf life. Thus, future research considering perishable scenarios is needed to develop effective policies in sectors such as food distribution or healthcare, where both service differentiation and spoilage risk are critical concerns. From a modelling perspective, however, incorporating perishability adds significant complexity, as it requires tracking the age distribution of inventory units over time. This additional state dimension increases the computational burden and may need new solution approaches or approximations to ensure tractability in realistic, multi-class inventory systems.

4.2. Theoretical and practical implications

This review offers several contributions to the theoretical and practical understanding of inventory rationing. The main theoretical contribution of this review is its refined classification framework for rationing policies. By explicitly distinguishing between time-dependent (dynamic) and state-dependent policies, our work provides a clearer conceptual lens for understanding the fundamental impact of lead times on policy optimality. This framework enables future researchers to more precisely position their contributions. Furthermore, the review highlights a consistent trade-off within the literature, where studies deriving optimal policies often rely on simplifying assumptions to maintain analytical tractability.

This shows the growing complexity of inventory rationing problems and the computational limits of traditional exact methods, evidenced by the common practice of state-space truncation in many studies. As researchers face even more intricate settings in the future, they will need to rely on advanced modelling techniques that can efficiently evaluate these (truncated) state spaces. To tackle such challenges, alternative methods for approximating the value function and optimal policy become essential. Our review shows that advanced modelling techniques designed to overcome the curse of dimensionality by approximating solutions are rarely applied in this research stream, which represents a significant research opportunity. For example, techniques from approximate dynamic programming (Powell 2011), such as state aggregation (see, e.g. Voelkel, Sachs, and Thonemann 2020) and basis function approximations (see, e.g. Wu, Evans, and Bae 2016), can reduce the effective dimensionality of the problem. Additionally, deep reinforcement learning approaches like actor-critic algorithms and proximal policy optimisation have proven effective in similar contexts (see, e.g. Goedhart, Haijema, and Akkerman 2023) by learning approximate policies that generalise across similar states.

The findings synthesised in this review also yield practical insights. The most direct insight is that implementing an inventory rationing policy provides significant performance benefits compared to a FIFO approach when serving customers with different priorities. Managers do not need to implement the most complex, theoretically optimal policies to achieve these gains; the literature shows that simpler static policies are not only easier to implement but also perform remarkably well, often achieving results very close to those of optimal ones.

While the choice of a specific policy structure can be flexible, a more fundamental decision for managers is how to model customer responses to stockouts. Our review shows that the distinction between modelling demand as backorders versus lost sales is a critical prerequisite, as the literature recommends different policy structures for each. Therefore, accurately understanding and modelling customer behaviour during a stockout is essential for designing an effective inventory strategy.

5. Conclusions

This study provided a structured literature review of 128 studies concerned with inventory rationing with critical level policies by building and extending on previous classifications in the field (see, e.g. de Kok et al. 2018; Teunter and Klein Haneveld 2008). With this review, we developed a structured classification of problem assumptions and characteristics to identify how different rationing policies are structured, based on the operating conditions of the systems. Since there is a significant difference between capacitated and non-capacitated systems (a production policy needs to be considered along with the inventory rationing policy for capacitated systems), we structured our review of inventory policies based on that differentiation. Furthermore, we extend the classification of rationing policy types by incorporating the category *state-dependent* into the already considered *static* and *dynamic* policies by previous attempts at structuring this research stream.

Our review shows that inventory rationing policies have been explored across a diverse range of problem settings. This body of research has established a strong foundation for understanding the impact of inventory rationing in contexts where product differentiation is required. However, extending prior work in this area presents several promising opportunities for future research, as we have discussed. Furthermore, a key motivation for the significant body of research on simpler, static policies is the perceived difficulty of implementing their more complex, theoretically optimal counterparts. This is a challenge that technologies like Industry 4.0 (Galati and Bigliardi 2019) and Logistics 4.0 (Winkelhaus and Grosse 2020) are poised to address. By lowering the implementation barriers for more sophisticated, optimal policies, the practical significance and theoretical depth of research in this area will only continue to grow.

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Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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Appendix A

Search strings used in the document title search:

TITLE-ABS-KEY ('inventory allocation' OR 'stock rationing' OR 'stock allocation' OR 'inventory rationing' OR 'rationing polic∗' OR 'rationing level∗' OR 'Critical level polic∗') AND (LIMIT-TO (SRCTYPE, 'j')) AND (LIMIT-TO (LANGUAGE, 'English'))

Appendix B

Table A1. Count of studies examining specific ordering and rationing policy combinations for non-capacitated settings.

Ordering policy review	Ordering policy type	Dynamic rationing policy	State-dependent rationing policy	Static rationing policy
Continuous	Base-stock		2	5
	s, Q	4		18
	s, S			1
	S-1, S		1	10
Periodic	R, Q	2		1
	R, S	11	5	7
	State-dependent		2	
Not applicable/ Not disclosed	Not applicable/Not disclosed	7	1	6

Table A2. Count of studies examining specific ordering and rationing policy combinations for capacitated settings.

Ordering policy review	Ordering policy type	Dynamic rationing policy	State- dependent rationing policy	Static rationing policy
Continuous	Base-stock		6	7
	Dynamic base-stock			1
	s, Q	1		
	s, S			2
	S-1, S			1
	State-dependent		25	
Periodic	R, Q		1	
	R, S		1	